

OZON Data Intelligence Pipeline

From Web Scrapping to Predictive Price Modeling

Project Report

Data Collection · Cleaning · EDA · ML Modeling

Team Member	GitHub	Primary Role
Bereket Kartbaev	Arab88	Scraping & Cleaning
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1. Introduction & Motivation

Ozon is one of Russia and Central Asia's largest e-commerce marketplaces, hosting millions of product listings across electronics, apparel, household goods, and more. The platform's scale makes it a rich source of real-world market data covering pricing, seller behavior, consumer ratings, and product specifications.

This project implements a **full data intelligence pipeline** on Ozon's consumer electronics segment — specifically **laptops, smartphones, and tablets**. Our motivation is threefold:

- Understand pricing structures and competitive dynamics in Kazakhstan's electronics market.
- Build a reproducible, end-to-end data engineering workflow from raw web data to machine-learning-ready features.
- Develop predictive models that can estimate product prices from hardware specifications, enabling smart purchasing decisions.

The pipeline follows five sequential stages — scraping, raw storage, cleaning, feature engineering, and modeling — mirroring industrial data science workflows and providing hands-on experience with anti-bot evasion, messy real-world data, and gradient boosting methods.

2. Data Source Description

2.1 Platform Overview

Data was collected from **ozon.ru**, a Russian-language marketplace operating across Russia, Kazakhstan, and Belarus. Product listings in three electronics categories were targeted: **laptops**, **smartphones**, and **tablets**.

2.2 Dataset Overview

16,311	5,376	5,382	5,553
Total products scraped	Laptops	Smartphones	Tablets

Category	Records	Engineered Features	Price Range (RUB)
Laptops	5,376	41	3,706 – 4,920,407
Smartphones	5,382	48	619 – 1,446,336
Tablets	5,553	53	1,413 – 1,728,329

2.3 Collected Fields

Each product card includes a rich set of attributes scraped directly from Ozon's internal pixis API:

Attribute Group	Fields
Pricing	Current price, original price, installment price, discount amount, discount %
Seller	Seller name, seller rating
Consumer	Card rating (avg review score), total review count
Hardware	CPU brand/model/GHz/cores, RAM, storage, GPU, screen size/matrix/Hz, battery
Product	OS, weight, dimensions, case material, connectivity features
Engineered	Discount %, RAM per core, pixel count, battery/gram, camera ratios

3. Scraping Methodology

3.1 Two-Stage Architecture

The scraping pipeline is split into two stages for fault tolerance and efficiency:

Stage 1 — Link Collection (`get_links.py`)

Uses undetected-chromedriver with Selenium to navigate Ozon category pages and harvest all product card URLs. Links are stored in JSON per category. This stage runs slowly (human-like delays) to avoid triggering bot detection.

Stage 2 — Card Parsing (`fetch_cards.py`)

Iterates through each collected URL and fetches product data by calling Ozon's internal pixis API directly — bypassing HTML parsing entirely. The API returns structured JSON with all product characteristics, which is far more reliable than scraping rendered HTML.

3.2 Anti-Bot Evasion Strategy

Ozon employs a sophisticated Cloudflare-backed anti-fraud stack that detects standard Selenium fingerprints (webdriver flags, automation properties, CDP artifacts). The following countermeasures were applied:

Technique	Description
undetected-chromedriver	Patches Chrome binary on the fly to remove webdriver flags
Real session cookies	Manual extraction of authenticated browser cookies via DevTools → Network → XHR → Copy as cURL
Real browser headers	User-Agent, Accept, Sec-Fetch headers mirroring an authentic browser session
Human-like delays	Random sleep intervals between requests to mimic natural browsing pace
Direct API calls	Bypasses rendered HTML and Cloudflare JS challenges by hitting the internal pixis endpoint

3.3 Data Storage

Raw data is versioned by pipeline stage under the `/data` directory:

- `data/raw/url/` — JSON files of product URLs per category
- `data/raw/categories/` — Full JSON responses from the pixis API (~19–22MB per category)
- `data/processing/not_fully/` — Intermediate CSVs during parsing
- `feature_engineering/data/cleaned/` — Final feature-engineered CSVs ready for modeling

4. Data Cleaning Steps & Justifications

The cleaning notebook (`notebooks/cleaning.ipynb`) applies a systematic multi-pass approach to transform raw API responses into analysis-ready datasets. Key steps are described below.

4.1 Duplicate Removal

Products appear multiple times across search result pages due to Ozon's recommendation engine re-inserting promoted items. Exact duplicates (same article ID) were dropped, keeping the first occurrence. This step was prioritized early to prevent inflated statistics in subsequent steps.

4.2 Type Coercion

Prices, ratings, RAM, and screen sizes arrived as strings with locale-specific formatting (e.g., '1 234,99 ■'). These were parsed to float/int. Boolean-like fields ('Yes'/'No') were mapped to 1/0 flags. Malformed values that could not be coerced were set to NaN for subsequent imputation.

4.3 Missing Value Handling

Continuous numeric features (RAM, battery, weight) were imputed with the category median to preserve distribution shape. Categorical fields (OS, CPU brand) that lacked a value were filled with the sentinel 'no_data' rather than dropping rows, preserving sample size. The smartphone dataset had only 5.8% missing product names — these rows were retained as pricing and specs were complete.

4.4 Outlier Treatment

Extreme price outliers (flagships like high-end Apple or gaming laptops) were retained rather than removed, as they represent genuine market segments. However, physically impossible values (e.g., 0 gram weight, negative prices) were nullified and treated as missing. This ensures the long right tail is modeled correctly.

4.5 Standardizing Category Labels

CPU names, brand strings, and OS labels contained many spelling variants from seller input. Regex-based normalization collapsed variants (e.g., 'Mediatek', 'MediaTek', 'media tek') to a canonical form. This reduced CPU brand cardinality by ~40% and improved downstream model encoding quality.

Result: After cleaning, the three datasets retained 5,376 / 5,382 / 5,553 records for laptops, smartphones, and tablets respectively — a retention rate above 95% from raw data. Missing-value rates dropped below 6% across all features.

5. Analysis Results & Visualizations

5.1 Price Distributions

Prices follow a heavy right-skewed distribution across all three categories. Smartphones and tablets cluster in the 50,000–100,000 RUB range, while laptops show a pronounced spike near 424,222 RUB (likely a popular flagship price point) with a long tail extending to ~5M RUB.

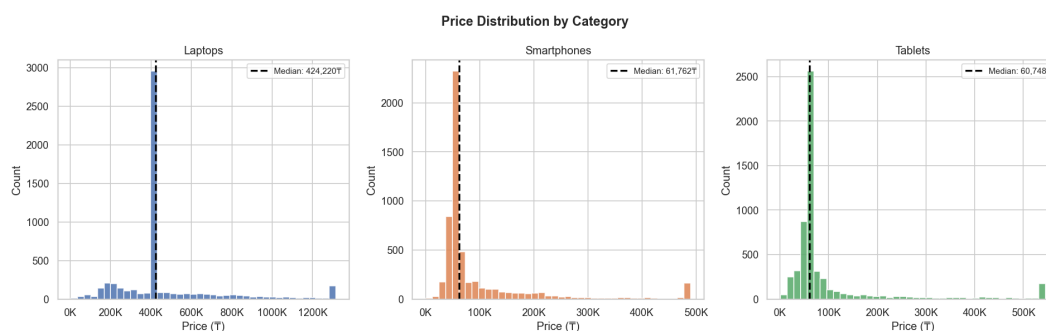


Figure 1. Price histogram across all three categories (log-transformed X axis).

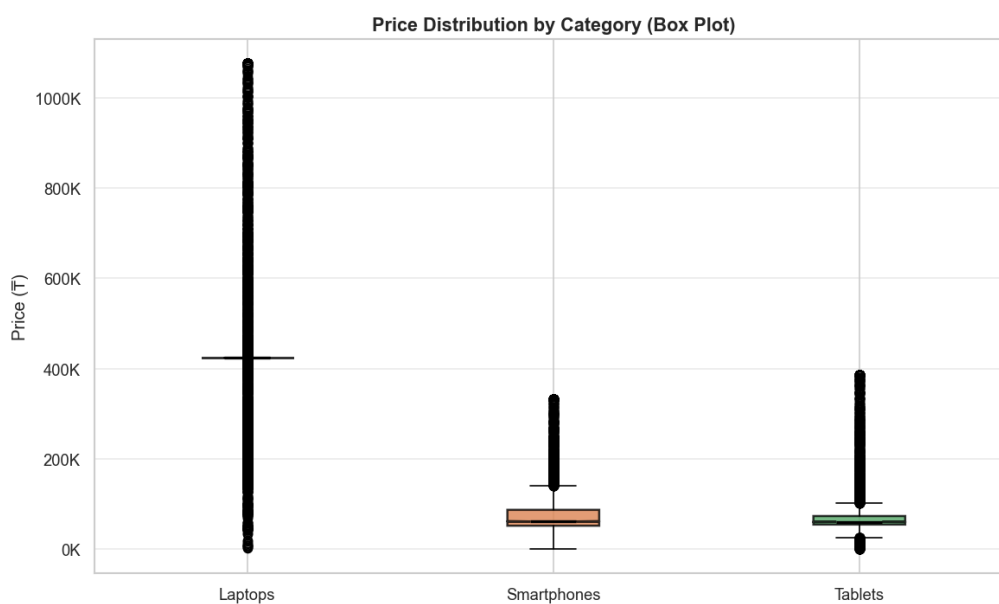


Figure 2. Boxplot comparison of price distributions — laptops show the widest spread.

5.2 Hardware Ecosystem

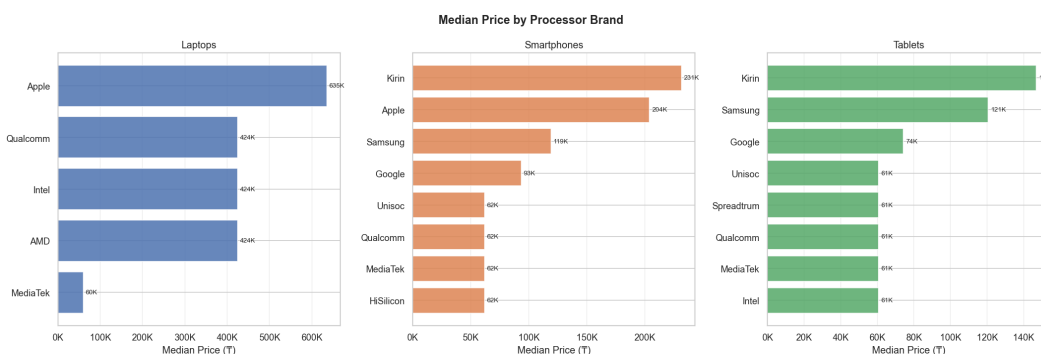


Figure 3. Laptop processor brand market share on Ozon. Intel dominates at 61%, AMD at 20%.

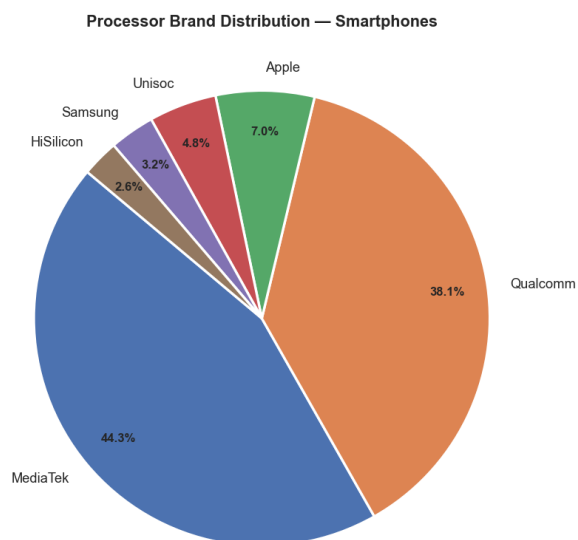


Figure 4. Smartphone processor pie — MediaTek leads (40%) followed closely by Qualcomm (34%).

5.3 OS Distribution

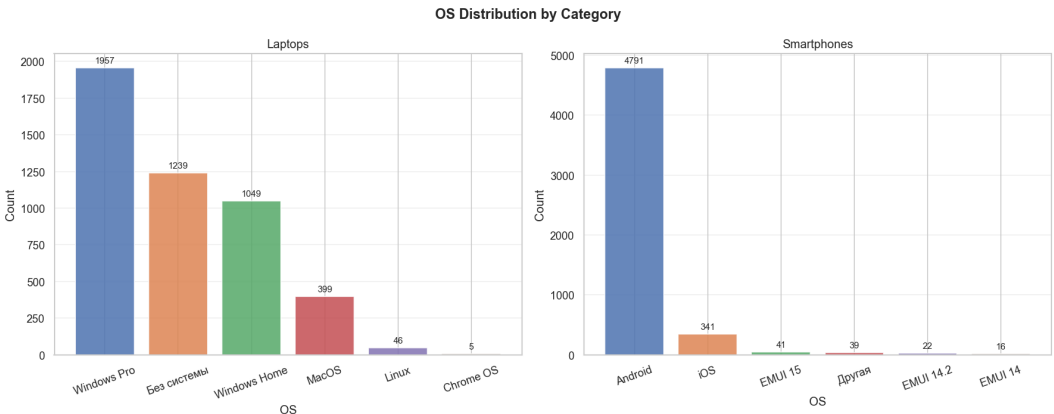


Figure 5. OS distribution across categories. Laptops: Windows Pro dominates; Smartphones: ~89% Android, ~6% iOS.

5.4 Feature Correlations — Laptops

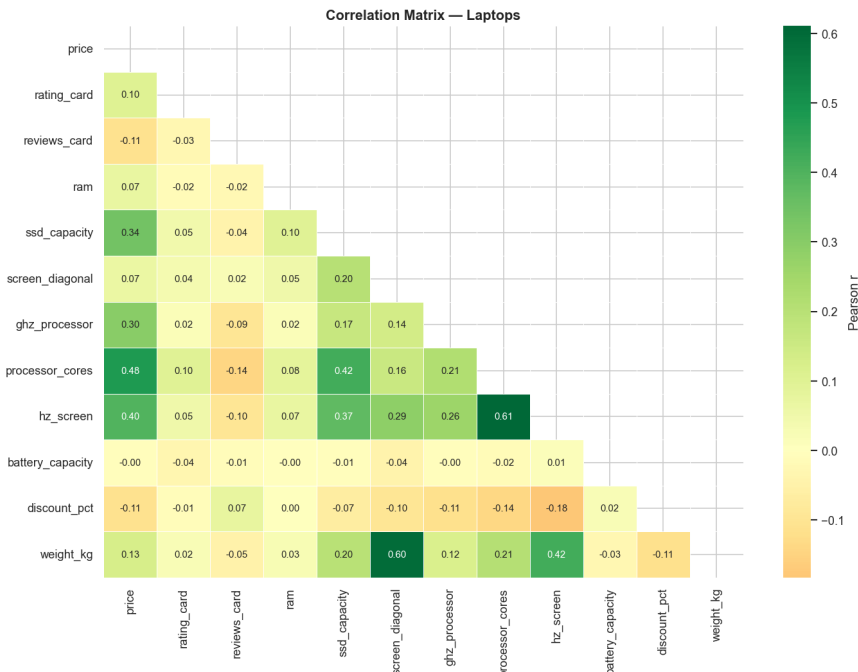


Figure 6. Correlation heatmap for laptop numerical features. Price correlates most strongly with RAM, screen size, and CPU power proxy.

5.5 RAM vs Price Relationship

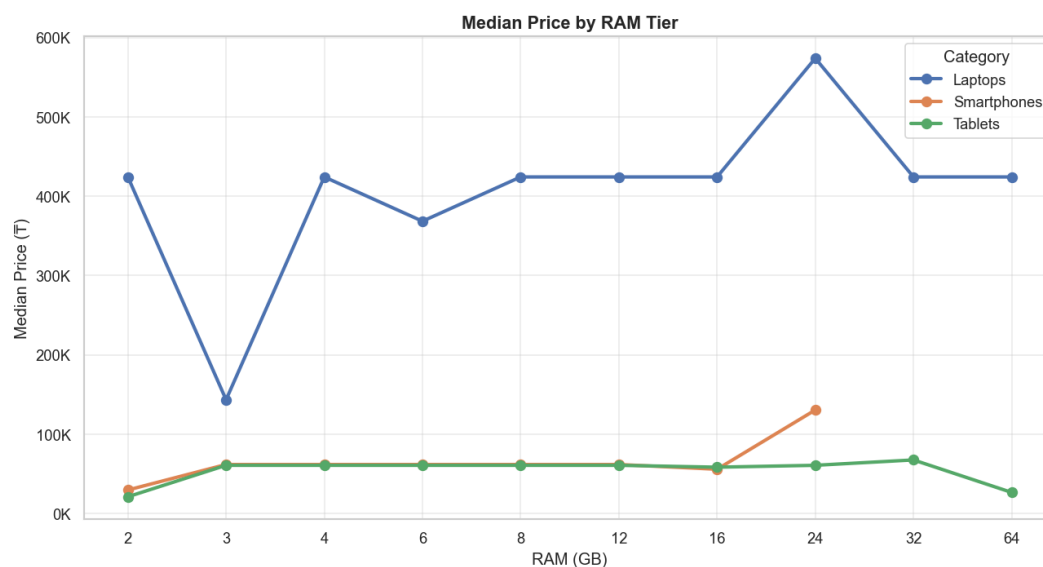


Figure 7. Median price by RAM tier. A strong monotonic relationship is visible — doubling RAM roughly doubles median price.

5.6 Rating Distribution

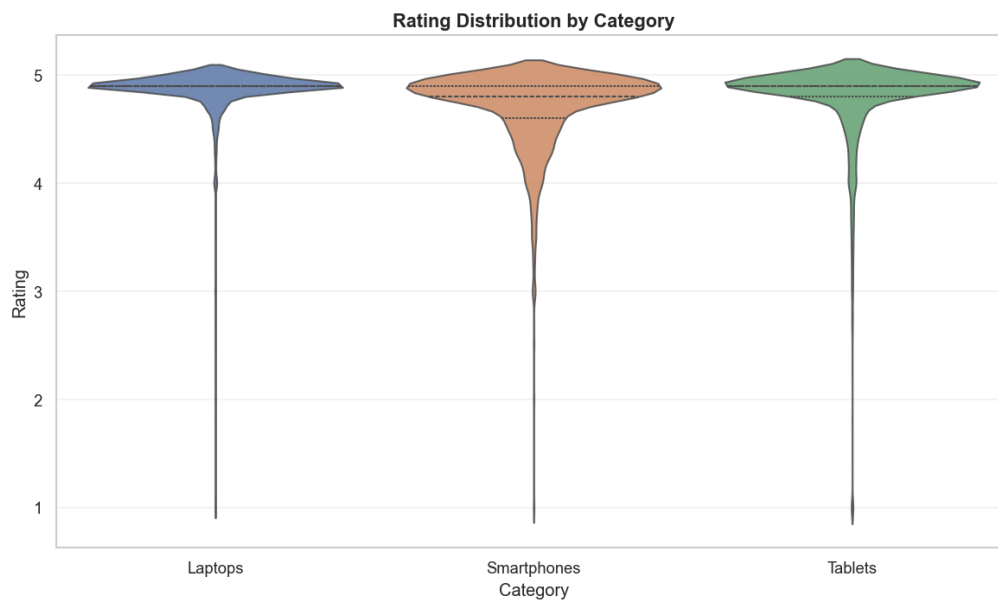


Figure 8. Violin plot of product ratings by category. All categories are left-skewed toward 5.0, with median ~4.7–4.9.

5.7 Smartphone Pairplot

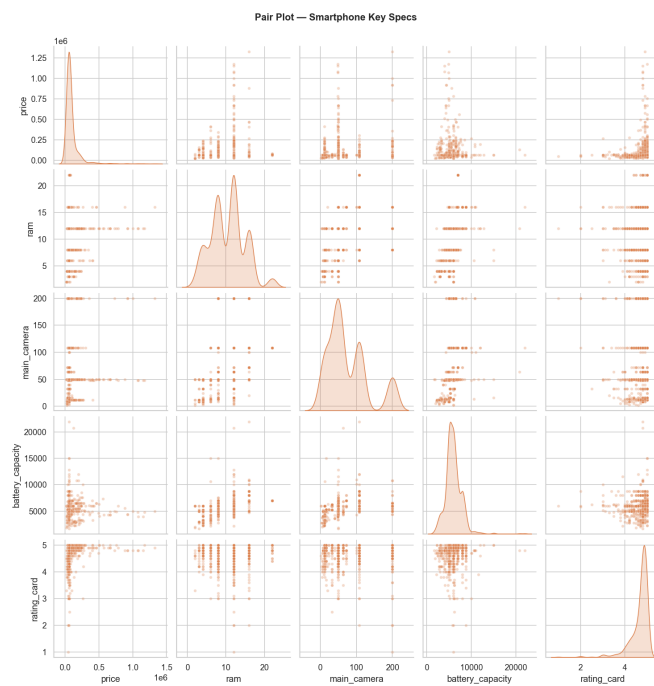


Figure 9. Pairplot of key smartphone features. Notable price–RAM cluster separation visible.

6. Key Findings & Insights

16,311	2.33M	~70%	4.7 / 5
Products collected	Total reviews (smartphones)	Avg listed discount (phones/tablets)	Average product rating

■ Pricing Segmentation

Smartphones and tablets occupy a budget-to-mid-range cluster (50–100K RUB) while laptops display trimodal pricing: budget (<100K), mid-range (300–500K), and premium (>1M). This suggests distinct buyer segments that should be modeled separately.

■ Discount Inflation

Average listed discounts of ~70% (smartphones/tablets) and ~50% (laptops) suggest widespread use of artificially inflated 'original prices' — a known dark pattern on Russian marketplaces. The discount_pct feature therefore has limited informational value as a genuine savings indicator.

■ Intel Dominance, MediaTek Rising

Intel captures 61% of laptop CPU market share on Ozon. In smartphones, MediaTek narrowly leads Qualcomm (40% vs 34%), reflecting the proliferation of affordable Android handsets in the CIS region. Apple's iOS accounts for only 6% of smartphones.

■ RAM is the Strongest Price Predictor for Laptops

The heatmap and pairplot analyses confirm that RAM size, storage, and CPU power proxy are the strongest numerical correlates of laptop price — consistent with consumer mental models that equate 'more RAM = better laptop'.

■ Rating Ceiling Effect

With median ratings of 4.7–4.9 out of 5, the marketplace exhibits a strong ceiling effect. Only products with below-average review counts fall below 4.5. This limits rating as a discriminating feature for recommendation systems.

■ Seller Quality Signal

Seller rating shows weak but positive correlation with product rating — premium sellers tend to list premium products. However, seller rating alone is insufficient for price prediction; product specs dominate.

7. Machine Learning Modeling

Price prediction models were built for all three categories using a two-model comparison strategy: a regularized linear baseline (Ridge) and a gradient boosting champion (CatBoost). All models predict log-transformed price to handle the right-skewed distribution, then exponentiate predictions back to RUB.

7.1 Model Results Summary

Category	Model	RMSE (log)	R ² (original scale)	Notes
Laptops	Ridge	0.267	0.443	Weak — non-linear effects
Laptops	CatBoost	0.378	0.542	Better on premium segment
Smartphones	Ridge	0.195	0.986	Excellent linear fit
Smartphones	CatBoost	—	—	Ridge sufficient
Tablets	Ridge	0.544	0.377	High variance in data
Tablets	CatBoost	0.403	0.740	Significant improvement

7.2 Key Modeling Observations

- **Smartphones** achieve near-perfect Ridge R² of 0.986, indicating that price is almost entirely determined by a linear combination of specs (RAM, storage, screen size, camera).
- **Laptops** benefit from CatBoost's ability to capture non-linear segment effects (gaming vs ultrabook vs workstation), boosting R² from 0.44 to 0.54.
- **Tablets** show the most improvement from CatBoost (0.38 → 0.74 R²), suggesting complex interaction effects between brand, OS, and connectivity specs.
- Log-price transformation was critical: without it, high-value outliers caused RMSE to be dominated by a handful of premium products.

7.3 Feature Importance

CatBoost's built-in feature importance analysis consistently ranked these features as top predictors across all three categories: RAM, CPU brand, storage capacity, screen diagonal, and OS. Engineered features (cpu_power_proxy, ram_per_price, pixel_count) provided meaningful additional signal — particularly for premium segment discrimination.

8. Challenges Faced

■ Anti-Bot Systems

Ozon employs Cloudflare with JavaScript fingerprinting. Standard Selenium was immediately blocked. The solution required patching Chrome with undetected-chromedriver, using real browser cookies extracted manually, and adding human-like request pacing. This required significant trial-and-error and was the most time-intensive challenge.

■ Inconsistent Seller-Provided Data

Product specifications are entered by sellers, not Ozon — resulting in wildly inconsistent formats (e.g., '8GB', '8■■', '8192 MB' for the same attribute). Normalization required extensive regex patterns and domain knowledge about plausible value ranges for each attribute.

■ Deeply Nested JSON Structure

Ozon's pixis API returns characteristics as nested arrays of key-value objects rather than flat JSON. Flattening this into a tabular format required recursive parsing logic and careful handling of optional/absent keys. Missing keys are category-specific (tablets have ~12 more attributes than laptops).

■ Price Outliers and Long Tails

The presence of ■5M+ laptops and ■1.4M+ smartphones creates extreme right-tail distributions that caused traditional regression metrics (RMSE) to be misleading. Log-price transformation and RMSE-on-log-scale evaluation were adopted to maintain interpretable model evaluation.

■ Memory and I/O Constraints

Raw JSON files reach 20MB each. Loading all three categories simultaneously exceeded notebook memory limits. Streaming JSON parsing and incremental CSV writing were adopted to process data in chunks without loading full datasets into RAM.

■ Discount Noise

Over 70% average discounts reflect artificial price inflation rather than genuine sales. Distinguishing 'real' from 'marketing' discounts requires external price history data (not available), which limited the utility of discount features in models.

9. Individual Contributions

The project was developed collaboratively by a four-person team with clearly delineated responsibilities tracked through GitHub commit history.

Bereket Kartbaev

GitHub: Arab88

Web Scraper & Data Engineer

- ✓ Designed and implemented the two-stage scraping architecture (get_links.py + fetch_cards.py)
- ✓ Implemented undetected-chromedriver setup and cookie-based anti-bot evasion
- ✓ Built the initial project repository structure and CI workflow
- ✓ Performed the initial data cleaning pass — deduplication, type coercion, and null handling
- ✓ Uploaded all raw data to version-controlled /data directories

Miras Kassymuly

GitHub: MirasKassymuly

Feature Engineer

- ✓ Designed and implemented feat_eng.ipynb for all three product categories
- ✓ Engineered derived features: discount_pct, cpu_power_proxy, ram_per_price, memory_per_price
- ✓ Parsed nested JSON characteristic arrays into flat tabular columns
- ✓ Created resolution features (res_w, res_h, pixel_count) from raw display strings
- ✓ Led the merge of feature engineering branch into main via pull request review

Raibek Kanatov

GitHub: Raibek

ML Engineer

- ✓ Built baseline Ridge regression models with log-price transformation for all three categories
- ✓ Trained and tuned CatBoost models using GridSearchCV and early stopping
- ✓ Implemented evaluation pipeline (RMSE log-scale + R^2 original-scale metrics)
- ✓ Corrected feature engineering outputs to ensure model-ready data formats
- ✓ Analyzed feature importances and documented model results

Sherkhan SD

GitHub: SherkhanSD

Data Analyst & Visualization

- ✓ Created all 9 exploratory visualizations (histograms, boxplots, heatmaps, pairplots, violin plots)
- ✓ Built the Power BI Dashboard (Dashboard.pbix) for interactive market exploration
- ✓ Conducted exploratory data analysis (EDA) on pricing, hardware specs, and OS distributions
- ✓ Authored the visualization notebook (notebooks/vizualization.ipynb)
- ✓ Identified and documented the discount inflation anomaly across all categories

10. Conclusions

This project successfully demonstrates a complete data intelligence pipeline on real-world e-commerce data from Ozon. Starting from raw web data protected by sophisticated anti-bot systems, the team built a reproducible workflow delivering 16,311 clean, feature-rich product records and competitive predictive models.

The key technical achievements include: robust anti-detection scraping via undetected-chromedriver and direct API access; systematic multi-pass data cleaning retaining >95% of records; comprehensive feature engineering adding 8–13 derived features per category; and price prediction models reaching R^2 of 0.986 for smartphones and 0.74 for tablets with CatBoost.

Future directions include tracking price changes over time to build temporal pricing models, expanding to additional categories (headphones, cameras), integrating seller history data for trust scoring, and deploying the best-performing models as a REST API for real-time price estimation.